

Project 2: The Server Commander

Provisioning a Linux Virtual Server and Hosting a Custom Webpage on AWS EC2

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Project	Cloud Computing Project 2
Cloud Provider Used:	AWS EC2 (Ubuntu Server 24.04 LTS)
Instance Public IP:	13.63.238.236

1. Objective

A startup is launching a new dynamic application and needs a dedicated server environment with full control over the operating system. This project demonstrates the process of provisioning a virtual machine in the cloud, securing remote access to it, installing web server software from the command line, and hosting a custom welcome page, simulating the responsibilities of a System Administrator.

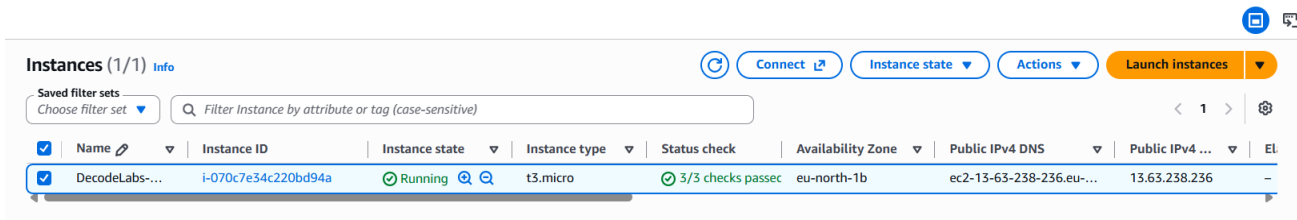
2. Tools & Services Used

- Cloud Platform: Amazon Web Services (AWS) – EC2
- Operating System: Ubuntu Server 24.04 LTS (HVM)
- Instance Type: t3.micro
- Web Server Software: Apache2 (or Nginx)
- Remote Access: SSH (OpenSSH client via PowerShell / Terminal)
- Local Machine: Windows (PowerShell)

3. Step 1 — Launch the Virtual Machine (EC2 Instance)

An Ubuntu Server 24.04 LTS instance (t3.micro) was launched via the AWS EC2 console with a new key pair (`decodeLabs-key.pem`) and a custom security group allowing SSH (port 22, restricted to my IP) and HTTP (port 80, open to the internet).

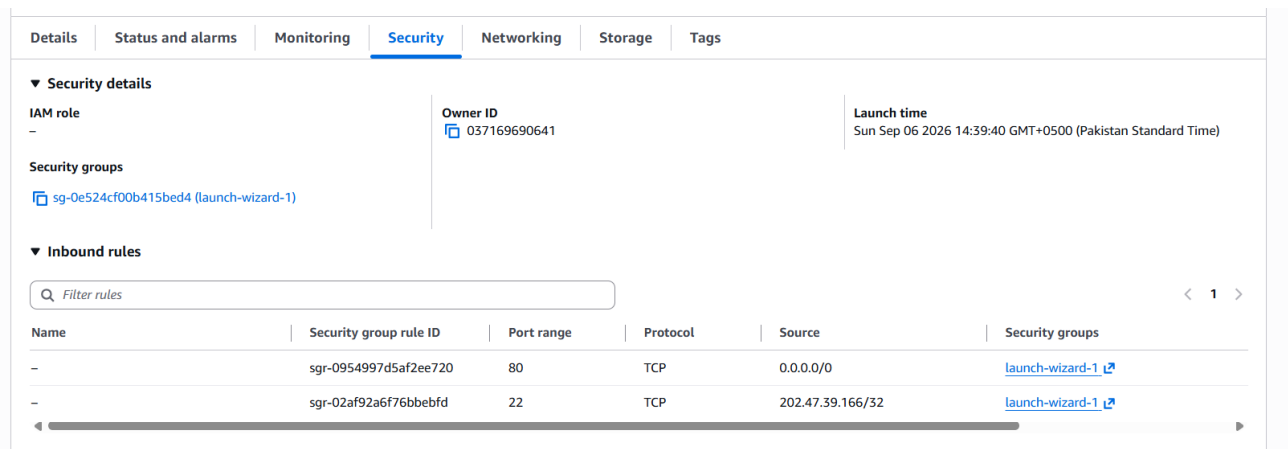
Screenshot 1: EC2 Instance Running



The screenshot shows the AWS Management Console 'Instances' page. A single instance named 'DecodeLabs-...' is listed with the ID 'i-070c7e34c220bd94a'. Its state is 'Running', and it is of type 't3.micro'. The instance is located in the 'eu-north-1' availability zone. The status check shows '3/3 checks passed'. The public IP address is '13.63.238.236'.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	EL
DecodeLabs-...	i-070c7e34c220bd94a	Running	t3.micro	3/3 checks passed	eu-north-1b	ec2-13-63-238-236.eu-...	13.63.238.236	-

Screenshot 2: Security Group Inbound Rules



The screenshot shows the 'Security' tab for a security group named 'sg-0e524cf00b415bed4 (launch-wizard-1)'. The 'Inbound rules' section is expanded, showing two rules. The first rule allows TCP traffic on port 80 from source '0.0.0.0/0'. The second rule allows TCP traffic on port 22 from source '202.47.39.166/32'.

Name	Security group rule ID	Port range	Protocol	Source	Security groups
-	sgr-0954997d5af2ee720	80	TCP	0.0.0.0/0	launch-wizard-1
-	sgr-02af92a6f76bbebfd	22	TCP	202.47.39.166/32	launch-wizard-1

4. Step 2 — Connect Securely via SSH

Connected to the instance from a local PowerShell terminal using the downloaded private key. File permissions on the .pem key were restricted using `icacls` so that only the current user account could read it, satisfying SSH's private key security requirement.

```
PS D:\> ssh -i "decodeLabs-key.pem" ubuntu@13.63.238.236 The
authenticity of host '13.63.238.236' can't be established.
Are you sure you want to continue connecting (yes/no)? yes Warning:
Permanently added '13.63.238.236' to the list of known hosts. Welcome
to Ubuntu 24.04 LTS ...
ubuntu@ip-172-31-xx-xx:~$
```

Screenshot 3: Successful SSH Login

```
D:\>ssh -i "decodeLabs-key.pem" ubuntu@13.63.238.236
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1017-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 15 10:50:50 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   39.9% of 6.71GB Processes:      117
Memory usage: 32%          Users logged in: 1
Swap usage:   0%           IPv4 address for ens5: 172.31.35.187

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ubuntu@ip-172-31-35-187:~$
```

5. Step 3 — Install a Web Server

Once connected, the package index was updated and Apache2 was installed directly from the command line.

```
ubuntu@ip-172-31-xx-xx:~$ sudo apt update && sudo apt upgrade -y
ubuntu@ip-172-31-xx-xx:~$ sudo apt install apache2 -y ubuntu@ip-172-31-xx-xx:~$ sudo systemctl status apache2
```

```
● apache2.service - The Apache HTTP Server
Active: active (running)
```

Screenshot 4: Apache Installation & Status

```
ubuntu@ip-172-31-35-187:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-15 09:58:47 UTC; 2h 4min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 50835 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 50838 (apache2)
    Tasks: 55 (limit: 1013)
   Memory: 6.1M (peak: 6.4M)
      CPU: 388ms
   CGroup: /system.slice/apache2.service
           └─50838 /usr/sbin/apache2 -k start
             └─50840 /usr/sbin/apache2 -k start
               └─50841 /usr/sbin/apache2 -k start

Sep 15 09:58:47 ip-172-31-35-187 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 15 09:58:47 ip-172-31-35-187 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

6. Step 4 — Host a Custom Welcome Page

The default Apache landing page was replaced with a custom HTML file at `/var/www/html/index.html`, edited directly on the server using nano.

```
ubuntu@ip-172-31-xx-xx:~$ sudo nano /var/www/html/index.html

<!DOCTYPE html>
<html>
<head>
  <title>DecodeLabs</title>
</head>
<body style="text-align:center; font-family:sans-serif; margin-top:100px;">
  <h1>Welcome to DecodeLabs</h1>
  <p>This server was
provisioned and configured manually via SSH.</p>
</body>
</html>
```

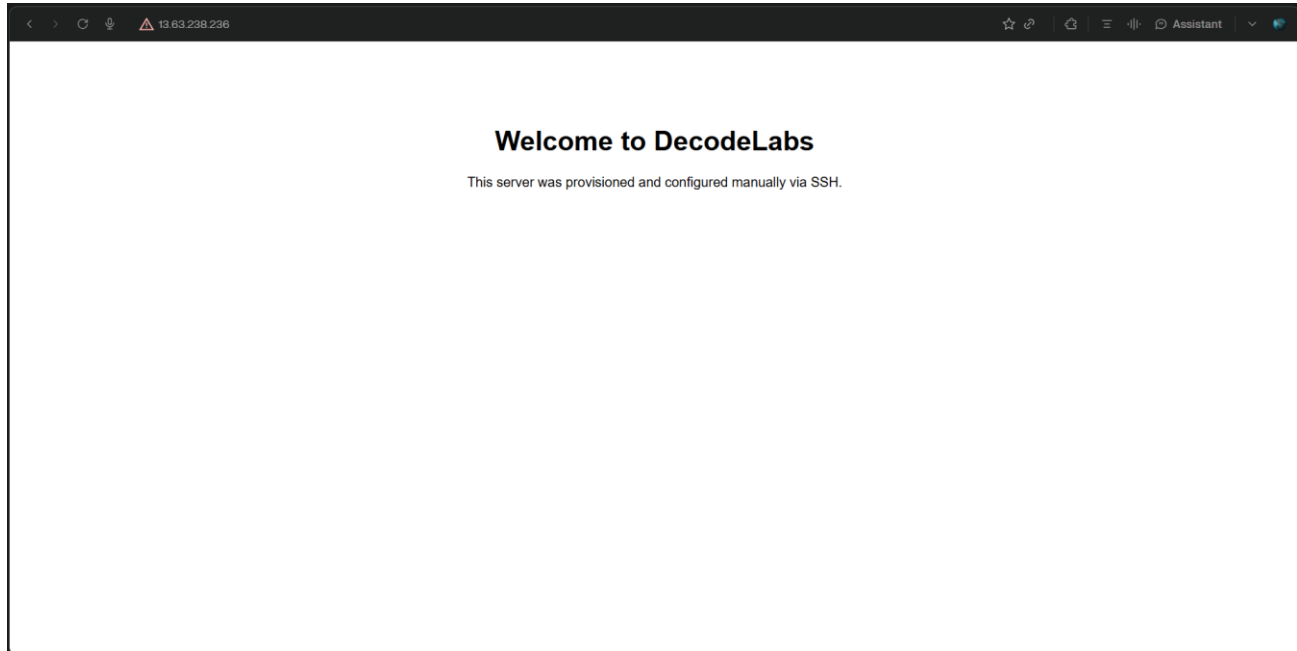
Screenshot 5: Editing index.html on the Server

```
GNU nano 7.2 /var/www/html/index.html *
<!DOCTYPE html>
<html>
<head>
  <title>DecodeLabs</title>
</head>
<body style="text-align:center; font-family:sans-serif; margin-top:100px;">
  <h1>Welcome to DecodeLabs</h1>
  <p>This server was provisioned and configured manually via SSH.</p>
</body>
</html>
```

7. Step 5 — Verify the Live Webpage

The public IPv4 address of the instance was opened in a web browser to confirm the custom page was being served correctly over HTTP.

Screenshot 6: Live Webpage in Browser



8. Conclusion

This project successfully demonstrated the end-to-end responsibilities of a System Administrator provisioning cloud infrastructure: launching a Linux virtual machine on AWS EC2, securing it with a restricted SSH key pair and firewall rules, connecting remotely via SSH, installing web server software from the command line, and deploying a custom webpage, all without relying on any managed or pre-configured hosting platform.